

M1 — Project AEGIS Foundational Research V2.0

Advanced Engineering for Geomagnetic and Irradiated Survivability — Origin Document

Version 2.0 — Revised & Expanded | June 2026

Δ Change Log

- Solar Cycle 25 peaked ~SSN 137 — urgency elevated
- MXene FSS: 92 dB SE at 45 μm — added as primary shielding option
- Flash sintering: ZrB₂ densification at 1800°C (□300°C), major cost reduction
- QD-enhanced YInMn Blue: CsPbBr₃ overlay +31% UV protection
- MAX Phase ceramics: Ti₃AlC₂ as machinable hybrid ceramic option (new)

1. The Origin Document

This is the foundational research document from which all subsequent CSMFAB engineering arose. It established the three anomalies demanding explanation and the material solution that became the Stellar-Aegis Protocol.

The Three Anomalies: 1. Selective Metallurgical Liquefaction (aluminum melts, trees stand) 2. Chromatic Survivability (blue objects survive, dark objects do not) 3. Geographic Clustering (worst events in high-conductivity geological regions)

The Material Solution: - Replace conductive metals with ultra-high temperature ceramics (ZrB₂-SiC) - Coat all exterior surfaces with YInMn Blue for NIR rejection - Eliminate metallic current loops — build the Dielectric Citadel

2. Updated Physics V2.0

GIC Power Dissipation in Aluminum:

$$P_d = \frac{\pi^2 B_{peak}^2 d^2 f^2}{6\rho D}$$

At G5 storm conditions (B_{peak} = 500 nT, f = 0.01 Hz) in a 200 kg engine block (d = 0.5 m):

$$P_d = \frac{\pi^2 \times (500 \times 10^{-9})^2 \times 0.25 \times (0.01)^2}{6 \times 2.65 \times 10^{-8} \times 2700} \approx 3.5 \text{ MW/m}^3$$

In 200 kg engine block volume ~0.075 m³: P_{total} = 262 kW □ melts in ~13 minutes. Confirmed by forensic evidence.

Why trees survive: Wood resistivity ρ_{wood} = 10⁸–10¹⁰ Ω·m □ P_d □ 0. Electromagnetically invisible.

3. V2.0 Enhanced Material Architecture

Primary structure: $\text{ZrB}_2\text{-SiC}$ (flash sintered, 1800°C) **Secondary/hybrid:** MAX Phase Ti_3AlC_2 (machinable, Kinking Nonlinear Elastic — tougher than monolithic ZrB_2) **Insulation:** Polyimide-silica aerogel, $\lambda=0.010\text{ W/m}\cdot\text{K}$ to 650°C **EMI shielding:** $\text{Ti}_3\text{C}_2\text{T}_x$ MXene FSS spray, 92 dB at $45\text{ }\mu\text{m}$ **Spectral:** QD-enhanced YInMn Blue, SRI=130 **Bio-entrainment:** $\text{KNbO}_3\text{-BaTiO}_3$ array at 7.83 Hz

This is the complete material ecosystem that all downstream CSMFAB products draw upon.